Seung Hyun Lee

Contact Information

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Summary

Ph.D. candidate in Computer Science and Engineering at the University of Michigan, advised by Prof. Stella X. Yu. I develop structured models for visual and embodied intelligence, spanning structure-aware perception, generative modeling, and robot learning. My work has appeared at CoRL, CVPR, ECCV, and ICCV, with research experience at Google DeepMind and Google Research.

Education

Ph.D. in Computer Science and Engineering, University of Michigan — Aug 2024 – May 2028
Research Advisor: Stella X. Yu

M.S. in Artificial Intelligence, Korea University — Mar 2022 – Aug 2024
Collaborated with Google Research and NVIDIA Research

B.S. in Computer Science, University of Seoul — Mar 2016 – Feb 2022
2 years of absence for obligatory military service
Undergraduate Research Assistant, Medical AI, Yonsei University Severance Hospital CCIDS

Research Experience

Google DeepMind, Remote — May 2026 – Sep 2026
Student Researcher

- Conducting research on embodied intelligence and robot learning.

University of Michigan, Ann Arbor, MI — Aug 2024 – Present
Ph.D. Researcher (Advisor: Prof. Stella X. Yu)

- Developed PRISM, a structured policy representation for robot learning (**CoRL 2026**).
- Developed SHED, a hierarchical visual representation for structure-aware dense prediction.

Google Research, Mountain View, CA — Jul 2023 – Mar 2024
Student Researcher

- Led **Parrot**: Pareto-optimal multi-reward RL framework for controllable text-to-image generation (**ECCV 2024 Oral**).
- Led **Cropper**: Vision-language model for in-context image cropping (**CVPR 2025**).
- Collaborated with Feng Yang, Irfan Essa, Ming-Hsuan Yang, and Yinxiao Li.

Conference Publications
(* Equal contribution)

Seung Hyun Lee, Stella X. Yu, PRISM: Polynomial Representations for Interaction-Structured Motor Control, In ***CoRL***, 2026 (Accepted).

Seung Hyun Lee*, Jijun Jiang*, Yiran Xu*, Zhuofang Li*, Junjie Ke, Yinxiao Li, Junfeng He, Steven Hickson, Katie Datsenko, Sangpil Kim, Ming-Hsuan Yang, Irfan Essa, Feng Yang, Cropper: Vision-Language Model for Image Cropping through In-Context Learning, In ***CVPR***, 2025, ***Google Research internship***.

Seung Hyun Lee, Yinxiao Li, Junjie Ke, Innfarn Yoo, Han Zhang, Jiahui Yu, Qifei Wang, Fei Deng, Glenn Entis, Junfeng He, Gang Li, Sangpil Kim, Irfan Essa, Feng Yang, Parrot: Pareto-optimal Multi-Reward Reinforcement Learning Framework for Text-to-Image Generation, In ***ECCV* 2024, (Oral) (< 2.3% acceptance rate),** ***Google Research internship***.

Yujin Jeong, Wonjeong Ryoo, **Seung Hyun Lee**, Dabin Seo, Wonmin Byeon, Sangpil Kim, Jinkyu Kim, The Power of Sound (TPoS): Audio Reactive Video Generation with Stable Diffusion, In ***ICCV***, 2023.

Wonseok Roh, **Seung Hyun Lee**, Won Jeong Ryoo, Gyeongrok Oh, Jakyung Lee, Soo Yeon Hwang, Hyung-gun Chi, Sangpil Kim, Functional Hand Type Prior for 3D Hand Pose Estimation and Action Recognition from Egocentric View Monocular Videos, In ***BMVC***, 2023, **Oral** (<5% acceptance rate).

Seung Hyun Lee, Wonseok Roh, Wonmin Byeon, Sang Ho Yoon, Chanyoung Kim, Jinkyu Kim*, Sangpil Kim*, Sound-Guided Semantic Image Manipulation, In ***CVPR***, 2022.

Seung Hyun Lee, Gyeongrok Oh, Wonmin Byeon, Chanyoung Kim, Won Jeong Ryoo, Sang Ho Yoon, Hyunjun Cho, Jihyun Bae, Jinkyu Kim*, Sangpil Kim*, Sound-Guided Semantic Video Generation, In ***ECCV***, 2022.

Journal Publications
(* Equal contribution)

Seung Hyun Lee*, Hyung-gun Chi*, Gyeongrok Oh, Wonmin Byeon, Sang Ho Yoon, Hyunje Park, Wonjun Cho, Jinkyu Kim*, Sangpil Kim*, Robust Sound-Guided Image Manipulation, ***Neural Networks***, 2024.

Seung Hyun Lee*, Sieun Kim*, Wonmin Byeon, Gyeongrok Oh, Sumin In, Hyeongcheol Park, Sang Ho Yoon, Sung-hee Hong, Jinkyu Kim, Sangpil Kim, Audio-Guided Implicit Neural Representation for Local Image Stylization, ***Computational Visual Media***, 2024.

Minjoon Jung, **Seung Hyun Lee**, Eun Seon Sim, Min Ho Jo, Yu Jin Lee, Hye Bin Choi, Junseok Kwon, Stagemix video generation using face and body keypoints detection, Multimedia Tools and Applications, 2022.

Workshop Publications

Seung Hyun Lee, Sang Ho Yoon, Jinkyu Kim*, Sangpil Kim*. Sound-Guided Semantic Image Manipulation. In ***NIPSW***–CtrlGen Workshop, 2021.

Seung Hyun Lee, Nahyuk Lee, Chanyoung Kim, Wonjeong Ryoo, Jinkyu Kim, Sangho Yoon, Sangpil Kim. Audio-Guided Image Manipulation for Artistic Paintings. In ***NIPSW***–Machine Learning for Creativity and Design Workshop, 2021.

Preprints

Seung Hyun Lee, Sangwoo Mo, Stella X. Yu, SHED Light on Segmentation for Dense Prediction, 2026.

Teaching Experience

University of Michigan, Ann Arbor, MI
- Lecturer, CSE 598: 3D Visual Representation — Fall 2026
- Lecturer/GSI, CSE 592: Foundations of Artificial Intelligence — Fall 2026
- Guest Lecturer, EECS 542: Depth Foundation Models — Winter 2026
- Graduate Student Instructor (GSI), EECS 442: Computer Vision — Winter 2026

Korea University, Seoul, Korea
- Teaching Assistant, Machine Learning — Mar 2023 – Jun 2023

Academic Service

Conference Reviewer: CoRL 2026; IJCAI 2025; ECCV 2024.

Journal Reviewer: Frontiers in Signal Processing, Image Processing.

Patents

Seung Hyun Lee. “Image Quality Improvement Apparatus Using Artificial Neural Network.” Korean Patent 10-2020-0021991, filed Feb. 2020; issued Sep. 2021.

Skills

Programming Languages: Python, C/C++, Java.

Tools: Deep learning frameworks (JAX, PyTorch, TensorFlow), Python libraries (Diffusers, OpenCV), and parallel programming (CUDA).

Honors and Selected Programs

Full Graduate Scholarship, Korea University — 2022 – 2024
Software Maestro, 11th Cohort (Selected Trainee), Korea — 2020

Leadership

President, Algorithm Club, University of Seoul — 2016 – 2018